

AMENDMENT NO. 2 MARCH 2017
TO
IS 16018 : 2012 WHEELED FIRE EXTINGUISHERS —
PERFORMANCE
AND CONSTRUCTION — SPECIFICATION

(Page 1, *clause 2*) — Insert the following new entry at an appropriate place:

<i>IS No.</i>	<i>Title</i>
4947 : 2006	Gas cartridges for use in fire extinguishers — Specification (<i>third revision</i>)

[Page 2, *clause 4.1* (d)] — Insert the following at the end:

- 'e) Foam-based;
- f) Water mist.'

(Page 3, *clause 5.2*) — Substitute the following for the existing clause:

‘5.2 Propellants

The propellants for stored pressure and cartridge-operated extinguishers shall be air, carbon dioxide and nitrogen. Gas cartridge shall conform to IS 4947 in all aspects including filling tolerance.'

[Page 3, *clause 5.4*] — Substitute the following for the existing clause:

‘5.4 Filling Tolerance

The actual charge of a wheeled fire extinguisher shall be nominal charge within the following limits:

- a) *Water-based extinguishers* — ± 5 percent by volume;
- b) *Powder extinguishers* — ± 2 percent by mass;
- c) *CO₂ extinguishers* — +0 , -5 percent by mass;
- d) *Clean agent extinguishers* — ± 5 percent by mass;

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- e) *Foam-based extinguishers* – ± 5 percent by volume; and
- a) *Water mist extinguishers* – ± 5 percent by volume.'

[Page 3, clause **5.5.1** (see also Amendment No.1)] — Substitute the following for the existing clause:

‘5.5.1 Nominal Charge

The following are the recommended charges for wheeled fire extinguishers:

- a) *Water-based extinguishers* – 10, 20, 30, 45, 60, 90 and 125 litre;
- b) *Powder-based extinguishers* – 25, 50 and 75 kg;
- c) *CO₂ based extinguishers* – 6.5, 9, 22.5 and 45 kg;
- d) *Clean agent extinguishers* – 10, 20, 30 and 50 kg;
- e) *Foam-based extinguishers* – 10, 20, 30, 45, 60, 90 and 125 litre; and
- f) *Water mist extinguishers* – 10, 20, 30, 45, 60, 90 and 125 litre.'

[Page 4, clause **6.3.1** (see also Amendment No. 1)] — Insert the following new clause at the end:

‘6.3.2 Test Procedure

Subject two extinguishers to the temperature cycles defined in Table 1, one extinguishers to each cycle.

Operate the extinguisher with the control valve fully open within 5 min of its removal from the conditioning chamber.

If it is not possible to operate the extinguisher within 5 min after its removal from the conditioning chamber, suitable means, such as insulation, shall be used to maintain the extinguisher at the conditioning temperature until discharge.'

[Page 5, clause **6.6.1**, para 1, line 2 (see also Amendment No. 1)] — Substitute ‘120 h’ for ‘5 days’.

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(Page 5, clause 6.6.1, para 2) — Substitute the following for the existing para:

‘One sample shall be tested.’

[Page 5, clause 6.6.2, Table 2, Sl No. (ii) and (iv), col (4)] — Substitute ‘ 27 ± 5 ’ for ‘ 27 ± 2 ’.

[Page 6, clause 7.2 (see also Amendment No. 1)] — Substitute the following for the existing clause:

‘7.2 Class A, Class B and Class C Fire Test

Wheeled fire extinguisher shall be tested for the following fire rating system as specified in IS 15683:

- a) Water-based extinguisher – 4A, *Min*
- b) Powder-based extinguisher
 - 1) ABC powder-based Extinguisher – 6A and 144 B, *Min*
 - 2) BC powder-based Extinguisher – 144 B, *Min*
- c) CO₂ based extinguisher – 55 B, *Min*
- d) Clean agent extinguisher – 3A and 89 B, *Min*
- e) Foam-based extinguisher – 4A and 144 B, *Min*
- f) Water mist extinguisher – 4A, *Min*

NOTE — The manufacturer shall declare fire ratings of their extinguisher. They may claim for one class or more than one class rating according to the extinguishing media. Testing and certification shall be based on the declaration and results obtained. If declared for Class ABC and in case sample fails either for Class A or Class B, it shall not be considered for certification.’

(Page 7, clause 8.1.6) — Insert the following new clause at the end:

‘8.1.7 A suitable drain plug be provided in wheeled fire extinguishers except for CO₂ and other gaseous extinguishers.’

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(Page 8, *clause 8.9.2*) — Substitute the following for the existing clause:

'8.9.2 Leakage Test (Type Test)

Each stored pressure fire extinguisher including valve assembly shall be dipped in suitable water tank filled with water, to the level more than the height of extinguisher and then a funnel with close narrow end made of transparent glass or transparent plastic filled with water, in inverted position, is used to cover the valve. Keep in this position for 12 h. There shall not be any collection of even single bubble by displacement of water on inside the tip of the inverted funnel after a lapse of 12 h.'

(Page 11, *clause 8.16.1.4*) — Substitute the following for the existing clause:

'8.16.1.4 The minimum indicated range of the pressure in the gauge shall be 0 to 20 bar. The gauge dial shall indicate, in green, the operable pressure range of the extinguisher. The zero, service, and maximum indicated gauge pressures shall be shown with suitable marks. The arc of the dial from the zero pressure point to the lower end of the operable range shall read 'Recharge' or be suitably marked. The arc of the dial from the higher end of the operable range to the maximum indicated pressure shall read 'Overcharged' or suitably marked. The area of service pressure range shall be marked with suitable colour. All numerals, letters, and characters in the recharge, operable, and overcharge portions of the dial shall be marked with any distinguished colour. Pointers shall be of prominent colour preferably in contrast to the base colour, and the tip of the pointer shall end in the arc.'

(Page 12, *clause 8.16.2*) — Substitute the following for the existing clause:

'8.16.2 Pressure Gauge Calibration Test

Calibrate the pressure gauge by standard method from an NABL accredited lab.'

(Page 13, *clause 8.16.8.1, line 5*) — Substitute ' $27 \pm 5\text{ }^{\circ}\text{C}$ ' for ' $27 \pm 2\text{ }^{\circ}\text{C}$ '.

(Page 14, *clause 8.19.2, para 2, line 4*) — Substitute ' $27 \pm 5\text{ }^{\circ}\text{C}$ ' for ' $27 \pm 2\text{ }^{\circ}\text{C}$ '.

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(Page 14, clause 9.1) — Insert the following at the end:

‘A small band of distinguishing colour of approximately 5 percent/prominent of surface area shall be painted for different type of extinguishers as given below:

<i>Extinguishers</i>	<i>Band Colour</i>
<i>Water-based extinguishers</i>	Red
<i>Powder-based extinguishers</i>	Blue
<i>CO₂ based extinguishers</i>	Black
<i>Clean agent extinguishers</i>	Green
<i>Foam-based extinguishers</i>	Yellow
<i>Water mist extinguishers</i>	Red